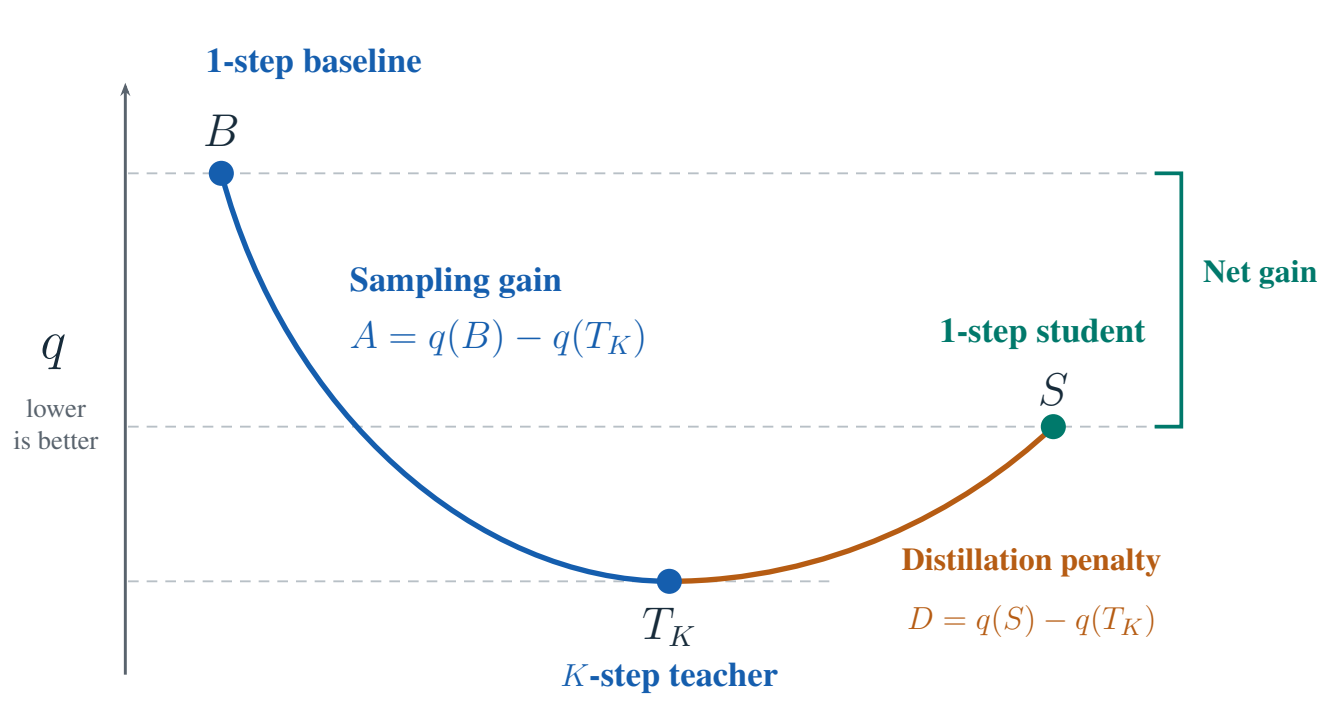

1-step baseline
B
Sampling gain
$A = q(B) - q(T_K)$
Net gain
1-step student
S
q
lower
is better
Distillation penalty
$D = q(S) - q(T_K)$
T_K
K-step teacher

$$q(B) - q(S) = A - D > 0$$